

APTITUDE

Calendar

- 0 → Sunday
- 1 → Monday
- ...
- 5 → Saturday
- 365 $\frac{1}{4}$ days
- ⇒ No. Y = 1 odd day
- ⇒ L.Y = 2 odd day
- ⇒ if Sanctuary 100/200/300 check $\frac{x}{400}$
- ⇒ 100 yrs → 5 odd days
- ⇒ 200 yrs → 3 " "
- ⇒ 300 yrs → 1 " "
- ⇒ 400 yrs → 0 " "

months

Jan → 3 odd days
Feb → 0/1
Mar → 3
April → 2
So on.

Calendar Repeation

- ⇒ L.Y repeats only L.Y
- ⇒ N.Y " " N.Y
- 2017, 2018, 2019, 2020, 2021, 22
- 1 1 1 2 1 1
- ∴ 2017 & 2023 repeats after 7 Count

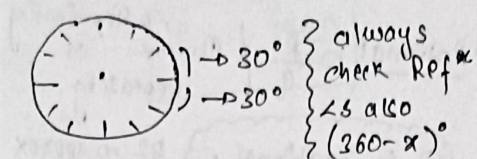
⇒ L.Y N.Y N.Y N.Y L.Y
28 6 11 11 28

⇒ last day of century cannot be 5, 3, 1, 0

Note:
1st Jan, 1992 → Wed
1st Jan, 2003 → ?
2+1+1+1+2+1+1+1+2+1+1 = 0
∴ Wed + 0 = Wed.

Clock

- ⇒ Relative speed M/H : 5.5°/min
- ⇒ H : 0.5°/min (moves)
- ⇒ M : 6°/min (moves)



⇒ Random time 7:23
7:00 → 210°
5.5° × 23 min → 126.5°
use diff technique from Std to Random
5.5 × minutes

⇒ Coincide / 0° : 12 hr = 11 times
24 hr = 22 "

⇒ 180° : 12 hr = 11 times
24 hr = 22 times

⇒ 90° : 12 hr = 22 times

⇒ excepts these 3 angles, all angles appears twice in every hr.

⇒ Angle to time :
2-3 { 90°
150° min
5.5 → 2 + 1

4-5 { 100°
20° 220°
5.5 5.5
Angle at which M will have to move from std time.
it can be Reflex. 340°

Average

- ⇒ Equal distribution.
- ⇒ Sum (n) : $\frac{n(n+1)}{2}$
- ⇒ Evens no's : (n+1) → avg.
- ⇒ n odds no's : n → avg.
- ⇒ n Even no's : n(n+1) → Sum
- ⇒ n odd no's : n(n) → Sum

Note
⇒ for average of n items : A
if x added ⇒ A decreases by a
then : $A - a(n+1) = x$

if x added ⇒ A increases by a
 $A + a(n+1) = x$

Percentage

- ⇒ Successive %
- 50% ↓ + 50% ↓ = 75% ↓
 $0.5 \times 0.5 = 0.25$
- 50% ↓ + 20% ↓ = 60% ↓
 $0.5 \times 0.8 = 0.4$
- 50% ↑ + 50% ↓ = 25% ↓
 $1.5 \times 0.5 = 0.75$
- 10% ↑ + 10% ↑ + 10% ↑ = 33.1% ↑
 $1.1 \times 1.1 \times 1.1 = 1.331$

⇒ Inverse proportion Logic.
rate ↑, time ↓ = A
⇒ A will be fixed, change other factors inversely

⇒ discount x% ⇒ $\square \times \frac{(100-x)}{100}$
decrement 20% = $\square \times 0.8$
⇒ increment 40% = $\square \times 1.40$

Time & Distance:

Ang $S = \frac{T.D}{T.P.}$ [timely distrib] can be diff.

$\frac{2D}{A.6} = \frac{D}{40} + \frac{D}{60}$

$\frac{2}{A.6} = \frac{1}{40} + \frac{1}{60}$

$\frac{2}{A.6} = \frac{1}{24}$

$A = 48$

$D = S \times T$ $\rightarrow$ 20% $\rightarrow$ $9t - 9t = \text{given}$

$60 \text{ kmph} \rightarrow 5 \text{ min early}$

$80 \text{ kmph} \rightarrow 10 \text{ min late}$

$\frac{D}{60} - \frac{D}{80} = 15$

$D = 60(t+10) = 80(t-5)$

meeting point: $D_1 + D_2 = T.D$

$S_1 \quad D_1 \quad D_2 \quad S_2$

Relative speed (opp) $= S_1 + S_2$

$\frac{T.D}{S_1 + S_2} = t \text{ of collision}$

Note

difference $L_1 - L_2 = \text{diff.}$

$D_1 + D_2 = T.D.$

at same time $\rightarrow$ relative speed

if train starts late by 2. $(t-2)$ will be consider.

crosses train + train: $2L = D$

" " + platform: $L + P = D$

" " + pole: $L = D$

Boat / upstream - Downstream:

$S_{\text{boat}} = \frac{U.S + D.S}{2}$

$S_{\text{water}} = \frac{D.S - U.S}{2}$

difference of time concept.

$\frac{x}{u+s} + \frac{x}{u-s} = P.t$

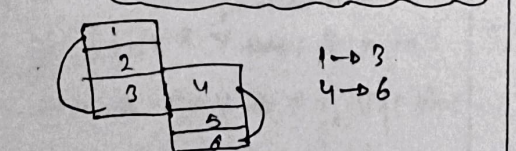
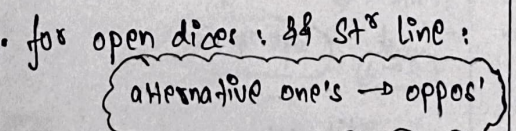
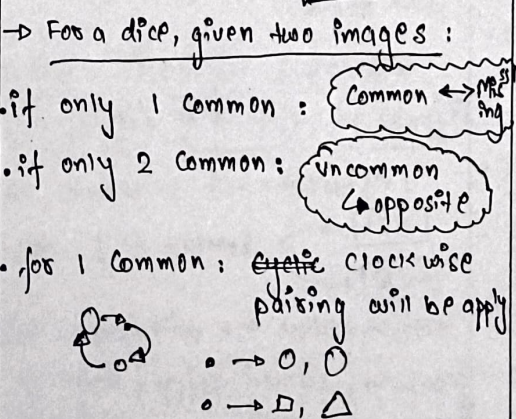
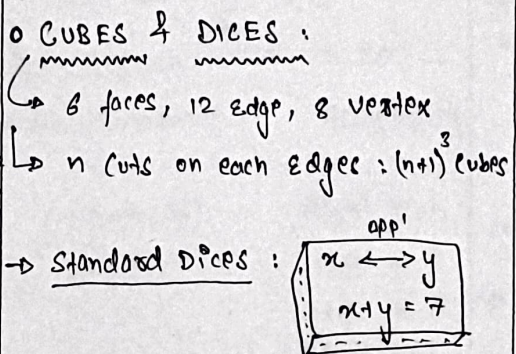
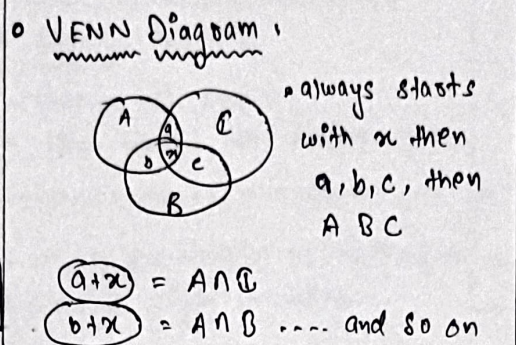
Race Comparison:

A beats B by 150 m of a Km race.

$\frac{A}{B} = \frac{1000}{850}$ info given is of last moment!

B beats C by 50 m

$\frac{B}{C} = \frac{1000}{950}$



Ranking & Arrangement:

if there is multiple arrangement, the answer will be (can't determined)

LPth & RPth of a given element:

No. of items = LP + RP - 1

$RP^{th} / LP^{th} = T - R/L + 1$

Coding & Decoding:

x means y

$\frac{x}{L} \rightarrow \text{code}$ $\frac{y}{L} \rightarrow \text{original}$

x is called $\frac{y}{L}$

$\frac{y}{L} \rightarrow \text{original}$ $\frac{L}{L} \rightarrow \text{code}$

A	E	J	O	T	Y
1	5	10	15	20	25

Relations:

Paternal: father's side

Maternal: mother's side

fore father > Great Grand.f > G

if Grandf: siblings

Cousins

